

# SETTING UP CONDA ENVIRONMENT

## 1. About Conda

### *What is Anaconda and Miniconda?*

- Anaconda is a full data science platform with Python, R, and many packages pre-installed.
- Miniconda is a lightweight version that installs only Python and the conda package manager, allowing users to install only required tools.

### Role in Bioinformatics

Conda ensures reproducibility, isolates environments, simplifies installation of complex tools, and allows workflows to run consistently across different systems.

## 2. Installation

### a. Windows Installation

- Open Command Prompt (search “cmd”)
- Type the following commands

```
curl https://repo.anaconda.com/miniconda/Miniconda3-latest-Windows-x86_64.exe -o .\miniconda.exe
start /wait "" .\miniconda.exe /S
del .\miniconda.exe
```

What these commands do:

- Downloads the miniconda software installer from the internet.
- Installation using the downloaded installer.
- Deletes installer since the miniconda is already installed.

## b. Mac Installation

Same as windows we will be installing miniconda using Mac terminal.

*Note: Since the Mac comes in two different kind of chipset (Intel and Apple Silicon) the installation will be a bit different.*

- Open Mac Terminal

### *For Intel Based Mac*

```
mkdir -p ~/miniconda3

curl https://repo.anaconda.com/miniconda/Miniconda3-latest-MacOSX-x84_64.sh \
-o ~/miniconda3/miniconda.sh

bash ~/miniconda3/miniconda.sh -b -u -p ~/miniconda3

rm ~/miniconda3/miniconda.sh

~/miniconda3/bin/conda init

source ~/.bashrc
```

### *For Apple Silicon Mac*

```
mkdir -p ~/miniconda3

curl https://repo.anaconda.com/miniconda/Miniconda3-latest-MacOSX-arm64.sh \
-o ~/miniconda3/miniconda.sh

bash ~/miniconda3/miniconda.sh -b -u -p ~/miniconda3

rm ~/miniconda3/miniconda.sh

~/miniconda3/bin/conda init

source ~/.bashrc
```

### c. Setting up Environment

- Restart your command prompt (Windows) or terminal(Mac)
- Check conda version

```
conda -version
```

- Activate conda base environment (Global Environment)

```
conda activate
```

## 3. Creating a Local Environment

It is important to have local environment inside the global environment to prevent clashes between Python/Tools/Softwares/Packages.

To create a local environment we use the following command

```
conda create --name myenv python=3.11
```

Here, you can personalize the name of the environment after the `--name` flag and also can specify the version of python you want to install or just leave it as `python` to install the latest stable version of python in your local environment.

At some point of creating conda environment you will be asked to press **[y]/[n]**, just type 'y' and press enter.

**Once installation is complete type the following commands:**

```
conda activate myenv #To enter your local environment
```

```
conda deactivate #To exit local environment and back to previous environment
```